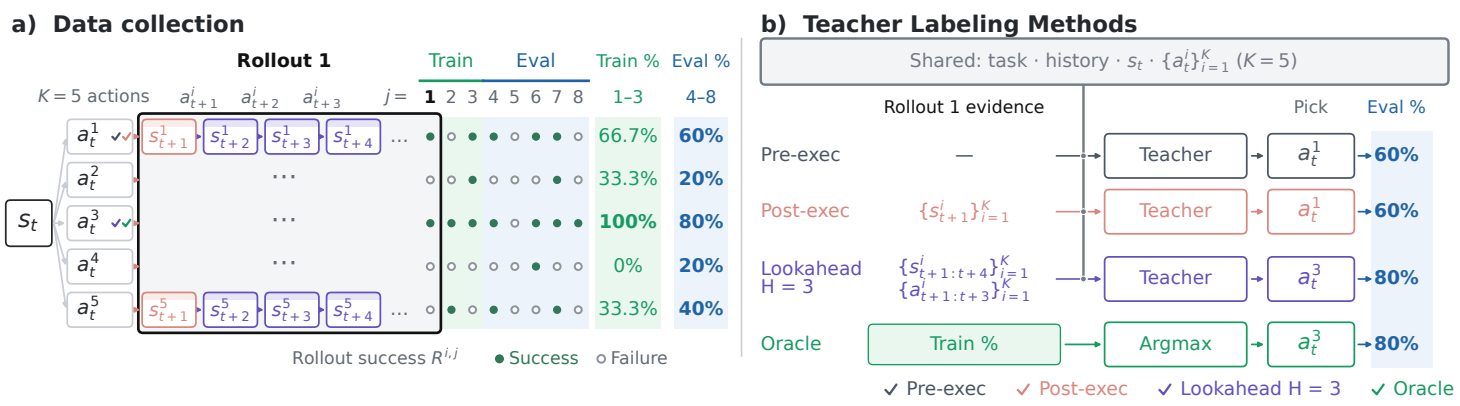
a) Data collection
Rollout 1
Train
Eval
Train %
Eval %
$K = 5$ actions
a^i_{t+1}
a^i_{t+2}
a^i_{t+3}
$j =$ 1 2 3 4 5 6 7 8
1–3
4–8
s_t
a_t^1
a_t^2
a_t^3
a_t^4
a_t^5
s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4}
s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4}
66.7%
33.3%
100%
0%
33.3%
60%
20%
80%
20%
40%
Rollout success $R^{i,j}$
Success
Failure
b) Teacher Labeling Methods
Shared: task · history · s_t · $\{a_t^i\}_{i=1}^K$ ($K = 5$)
Rollout 1 evidence
Pick
Eval %
Pre-exec
—
Post-exec
$\{s^i_{t+1}\}_{i=1}^K$
Lookahead H = 3
$\{s^i_{t+1:t+4}\}_{i=1}^K$
$\{a^i_{t+1:t+3}\}_{i=1}^K$
Oracle
Train %
Teacher
Argmax
a_t^1
a_t^3
60%
60%
80%
80%
✓ Pre-exec
✓ Post-exec
✓ Lookahead H = 3
✓ Oracle